size and distance from its star to sustain liquid water on its surface—a hint, though not a certain one, that life could find a way to thrive. In the trillion other galaxies visible across the observable universe, there could be countless other species, with their own civilizations, arts, cultures, and scientific endeavors, all telling their stories of the universe from their own perspectives, slowly discovering their own primordial past. On each of those worlds, creatures like, or unlike, ourselves might be detecting the faint hum of the cosmic microwave background, deducing the existence of the Big Bang and the staggering knowledge that our shared cosmos does not go back forever in time, but had a first moment, a first particle, a first star.

And those other beings, like us, might be coming to the same realization: that a universe that is not static, that had a distinct beginning, must also, inevitably, have an end.